

# Urban Wünsch

Assistant Professor, Chalmers University of Technology

Researcher working on organic compounds dissolved in water — disentangling the composition of complex environmental samples to understand the fate of Dissolved Organic Matter (DOM) in aquatic systems. As of late 2026: 34 publications (10 as lead), H-index 19, 2100+ citations.

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## Education

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### Technical University of Denmark

PhD 2018

### University of Applied Sciences Bremerhaven

M.Sc. Biotechnology 2014

### University of Applied Sciences Bingen

B.Sc. Biotechnology 2011

## Experience

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### Chalmers University of Technology, Architecture and Civil Engineering

Assistant Professor 2025 – Present

### DTU Aqua, National Institute for Aquatic Resources

Assistant Professor 2024 – 2025

Postdoc 2020 – 2024

### Chalmers University of Technology, Architecture and Civil Engineering

Postdoc 2018 – 2020

### Technical University of Denmark, National Institute for Aquatic Resources

PhD student 2014 – 2017

## Teaching

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### **Blue Bioeconomy Innovation**

2023 – 2024

Course responsible

M.Eng, ~40 students

### **Ocean science & technology**

2015 – 2025

Teaching topics in master-level courses

### **Machine learning for spectroscopic data analysis**

2018 – Present

Co-organizer of online course

For Master- and PhD-students

### **Introduction to Chemistry & Urban spaces and functions for civil engineers**

2018 –  
2020

Teaching, bachelor-level courses

Large classrooms

## Supervision

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### **Luis Salvatierra Chung Sang (PhD)**

2026 – Present

Co-supervisor

DTU Aqua (Denmark)

### **Julia Iwan (PhD)**

2025 – Present

Main supervisor

### **Frederik Lenk (B.Sc.)**

2024 – 2024

Main supervisor

Special course

### **Maria Papadimitraki (Research Assistant)**

2023 – 2023

Co-supervisor

University of Southern Denmark

### **Emilie Tage Andresen (B.Sc.)**

2022 – 2022

Co-supervisor

DTU Chemistry (Denmark)

## **Linea Gry Ebbesen (B.Eng.)**

2022 – 2022

Co-supervisor

DTU Environment (Denmark)

## **Laila Vinther & Cecilie Hjelm Hvas Hansen (M.Sc.)**

2021 – 2021

Co-supervisor, voluntary special course

DTU Environment (Denmark)

## **Evelyn Namuga Kasule (B.Sc.)**

2021 – 2021

Co-supervisor

DTU Environment (Denmark)

## **Signe Melbye Andersen (M.Sc.)**

2020 – 2022

Co-supervisor

DTU Aqua (Denmark)

## **Anders Dalhoff Bruhn Jensen (PhD)**

2019 – 2023

Co-supervisor

DTU Aqua (Denmark)

## **Xianyu Kong (PhD)**

2019 – 2022

PhD committee member

Alfred Wegener Institute (Germany)

## **Nashita Moona (PhD)**

2018 – 2021

Co-supervisor

Chalmers (Sweden)

## **Skills**

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### **Dissolved organic matter characterization**

Excitation-Emission Matrix (EEM) spectroscopy

Absorbance spectroscopy

High Performance Size Exclusion Chromatography  
(HPSEC)

Fourier Transform Ion Cyclotron Resonance Mass  
Spectrometry (FT-ICR-MS)

### **Chemometrics**

Principal Component Analysis (PCA)

Parallel Factor Analysis (PARAFAC & PARAFAC2)

N-way Partial Least Squares (N-PLS)

Advanced Coupled Matrix and Tensor Factorization  
(ACMTF)

## Research grants

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### **Dissolved Organic Matter Biogeochemistry - innovative methods and sensors** 2025

Strategic funding, Chalmers thematic area Ocean

2025-2029, Principal Investigator

### **A new approach to laccase discovery from the sea** 2023

Independent Research Fund Denmark

2.5 million DKK, named applicant

### **New eyes on old GAC** 2022

FORMAS, Sweden

146,000 SEK, co-applicant

### **Oxidation Demand and Rate (ODaR) as a sensitive proxy of biostability in drinking water** 2021

FORMAS, Sweden

26,000 SEK, co-applicant

### **Enhancing Biogeochemical Fingerprints of Natural Organic Matter with Data Fusion** 2019

Åforsk Foundation

400,000 SEK, main applicant

### **An optical sensor for reducing biocide dependence in chemical manufacturing** 2019

Vinnova, Swedish Governmental Agency for Innovation Systems

474,000 SEK, co-applicant

### **2-year postdoctoral scholarship** 2019

Carl Tryggers Stiftelse

552,000 SEK, main author

## Recent conferences

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### **Photosensitivity of fluorescent dissolved organic matter across the arctic ocean** 2023

Aquatic Sciences Meeting 2023, Mallorca, Spain

## **Combining Physical and Mathematical Chromatography to Decipher the Molecular Fingerprints of Dissolved Organic Matter**

2021

20th Conference of the International Humic Substances Society (online)

## **Mathematical chromatography deciphers the molecular fingerprints of dissolved organic matter**

2021

Aquatic Sciences Meeting 2021 (online)

## **Membership in Scientific Communities**

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### **Association for the Sciences of Limnology and Oceanography (ASLO)**

Member since 2014

### **International Humic Substances Society (IHSS)**

Member since 2018